

Caden (ZiHuan) Zhang

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EDUCATION

University of British Columbia

Bachelor of Science in Computer Science - GPA: 4.08/4.33 (94.3%)

Vancouver, BC

Sept. 2024 – April 2028

PROJECTS

Party Lab | C++, Raylib, Lua, Steam API

July 2024 – Dec. 2024

- Built a **C++** multiplayer game framework/hub with a custom declarative UI system using **Raylib**
- Implemented **client-server networking** with Valve's **GameNetworkingSockets API**, handling connection state and message dispatch; integrated the **Steam API** for friend invites and leaderboards
- Embedded **Lua** scripting via **sol2 library** in a sandboxed runtime, enabling modding and custom game modes without touching native engine code

Astrodescent | C#, Raylib, GLSL, Steam

Aug. 2026 – Present

- Shipping a 2D space-mining game on **Steam** built without a game engine: custom **deferred renderer**, post-processing pipeline, dynamic 2D lighting, texture atlasing, and a pixel-font UI system
- Wrote **gameplay physics** from scratch, including swept **AABB** tile collision and **DDA grid raycasting** for mining and tool targeting
- Designed a **data-driven content architecture** for tiles, items, biomes, and planets, with procedural generation using Perlin noise and Poisson-disc feature placement

Gunplay | Unity, C#

July 2025 – January 2026

- Contributed to the **UBC Game Development Team's** physically based, hyper-realistic **FPS**
- Implemented and tuned **ragdoll physics** for realistic character reactions and physical interactions
- Implemented the complete **UAV system**, including **aircraft movement**, and **player input handling**

Sand of Souls | Godot, GDScript, GLSL

July 2026

- Survivor shooter built in **4 days** with a **19-person team** for **GMTK Game Jam 2026** (10,600+ entries)
- Implemented the game's **visual effects** layer in **GLSL**, including enemy disintegration, screen-space vignette and desaturation, bloom, muzzle flashes, projectile trails, railgun charge, and etc
- Built enemy AIs with a wave-based difficulty scaler
- Implemented the full menu/HUD flow, including UI layout and animations

TECHNICAL WORK EXPERIENCE

Full Stack Developer — Checker Software

January 2026 – August 2026

- Developed a web-based **DOCX renderer** with **Rust**, speeding up document comparison by **60%**
- Optimized **LibreOffice WASM** for the web, lowering bundle size by **25%**
- Developed the **localization system** for the Diffchecker desktop app, enabling multi-language support

SWE Highschool Internship — Nokia

July 2023 – August 2023

- Improved stability and performance of the Netguard Endpoint Detection and Response product by identifying and resolving 3 recurring performance bottlenecks across **Linux VMs** using **Grafana** and **Prometheus**
- Increased test coverage by 10% through implementing **RED Team assessments** with **Caldera** and **Robot Framework**, enhancing product reliability across multiple platforms

TECHNICAL SKILLS

Languages (Advanced): C++, C#, Java

Languages (Intermediate): C, Python, JavaScript/TypeScript, Rust, GDScript, GLSL

Systems & Networking: multithreading, memory management, client-server architecture, GameNetworkingSockets

Frameworks: Unity, Godot, Raylib, NodeJS, React, Svelte, Vue, Tauri

Tools: Git, Github, Github Actions, Docker, Jira, Confluence, Trello